Q1)

(A) Explain Specialization and generalization with suitable Example.

→ (i) Specialization:

(a) Specialization is a top-down approach in which a higher-level entity is broken into two or more lower level entities based on distinct & specific characteristic

(b) Example: A higher-level entity account can be specialized into Saving_Account and Current_Account. Both inherit the base attributes of Account (like Account_Number) but Savings_Account might have a specific attribute like Interest_Rate, while Current_Account has Overdraft_Limit.

(ii) Generalization:-

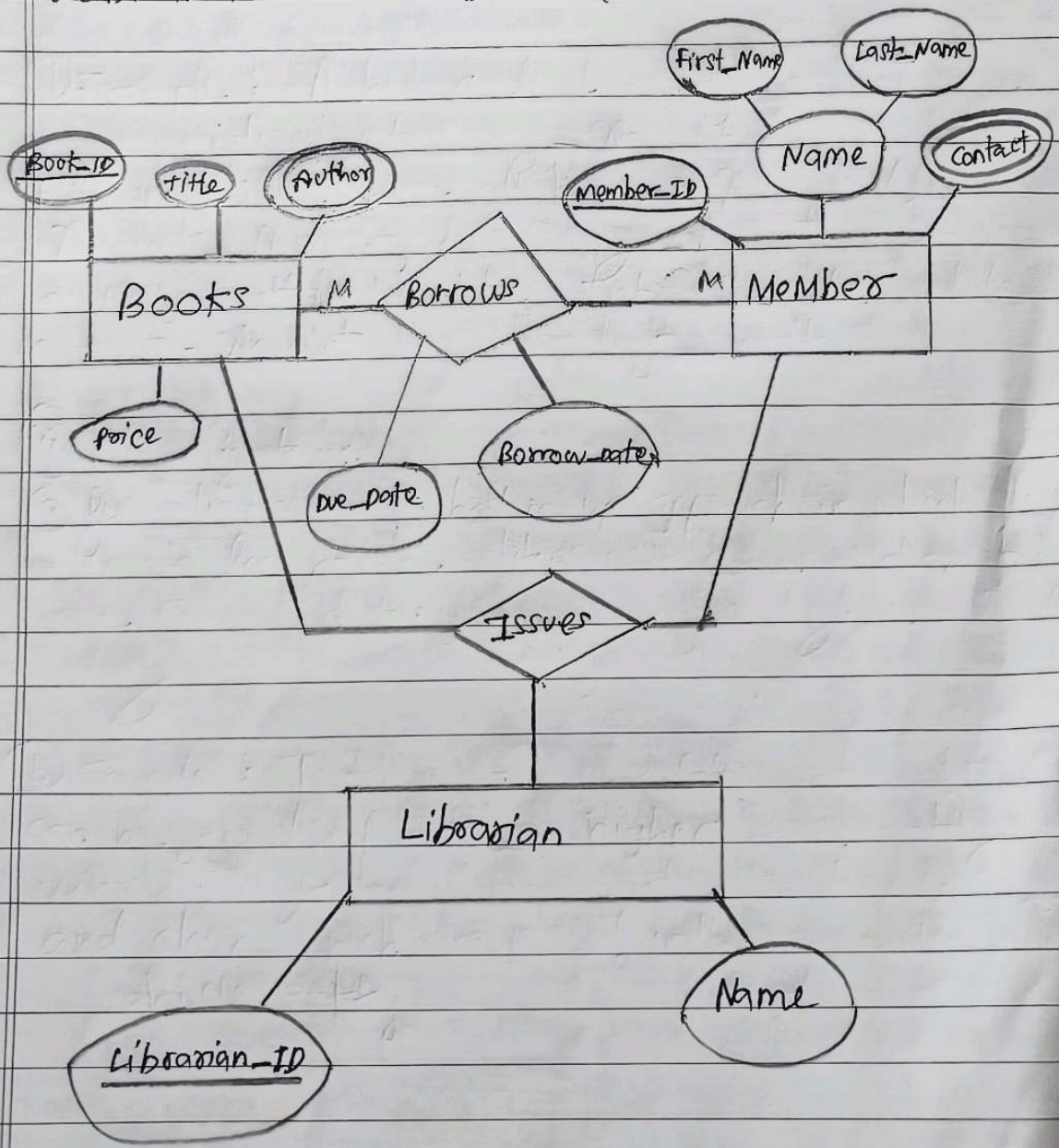
(a) Generalization is a bottom up approach in which two or more lower-level entities with common characteristics are combined to form a single higher-level entity

(b) Example: The lower-level entities car and Truck can be generalized into a higher-level entity called vehicle. They share common attributes like vehicle_ID and colour, which are placed in the generalized vehicle entity

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Q.1)

(B) Draw ER model for library Management System. A library maintains books, members and librarians. Each Book has Book_ID, title, Author and price and Each member has Member_ID, name and contact. A member can borrow many books. A book can be borrowed by different members over time. Each librarian issues book to members.



Q2)

(A) Explain Fundamentals operations of relational algebra with suitable Example.

(i) Select (σ)
(ii) Project (Π)
(iii) Union (∪)
(iv) Set difference (−)
(v) Cartesian product (X)

→ (i) Select (σ)

- purpose : Filters rows (tuples) from a relation that satisfy a specific condition
- Example : To find Employees earning more than 50,000

$\sigma_{Salary > 50,000}(Employees)$.

(ii) Project (Π)

- purpose : selects specific columns (attributes) from a relation and discards the rest automatically removing duplicate rows.
- Example : To Extract only the names and department of Employees

$\Pi_{name, department}(Employee)$

(iii) Union (∪)

- purpose : Combines all tuples from two union-compatible relations, removing duplicates. Both relations must have the same number of attributes and matching domains
- Example : To find all active and past student names

$\Pi_{Name}(Active_Students) \cup \Pi_{name}(Past_students)$

(iv) Set Difference (−)

- Purpose :- Returns tuples that are present in the first relation but completely absent from the second. Relations must be union compatible.
- Example :- To find customers who are depositors but not borrowers

$$\Pi_{Name}(Depositor) - \Pi_{Name}(Borrower)$$

(v) Cartesian product (×)

- Purpose : combines information from two relations by pairing every tuple of the first relation with every tuple of the second relation (cross join).
- Example :- To combine all student with all available courses :

  student × course.

Q2)

(B) consider the following relational database and construct the relational algebra queries for the following

Account (Branch-name, Account-name, Balance)
Loan (Branch-name, Loan-number, Amount)
Depositor (cust-name, Account number)
Borrower (cust-name, Loan-number)

→

(a) Find the names of customer who have an account at the bank.

$\Pi_{cust_name}(Depositor) \cap \Pi_{cust_name}(Borrower)$

(b) Find the customer names who have an account at roadbridge branch.

$\Pi_{cust_name}(\sigma_{Branch_name = Roadbridge}(Depositor \bowtie Account))$

(c) Find account balance at perryridge

$\Pi_{Balance}(\sigma_{Branch_name = Perryridge}(Account))$

(d) Find the customer names whose loan amount is greater than 50000

$\Pi_{cust_name}(\sigma_{amount > 50000}(Borrower \bowtie Loan))$

(e) Find a customer who have an account but not a loan in bank account

$\Pi_{cus_name}(Depositor) - \Pi_{cust_name}(Borrower)$